

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
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(1) Research Program/Project/Study Title: Utilization of Milkfish (<i>Chanos chanos</i>) Bone Flour in Making of Kroepoek	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Pontevedra
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☒ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Food/MilkFish	(8) Discipline/Field of Specialization Physical Science Food Technology Food Chemistry Hospitality Management
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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Milkfish (*Chanoschanos*) is an important food staple in the Philippines as it is one of the most widely consumed fish in the country. Other than its remarkable taste, many would recognize it for the large number of bones it contains. With this, deboning became a common practice provided by local fish sellers as a way to manage this inconvenience during consumption. This procedure makes milkfish bones one of the food waste at disposal in local markets. However, this neglects the potential of milkfish bones as a resource as milkfish farming in the Philippines has been increasing significantly these past few years with around 400 thousand MT(metric ton) produced annually based on the local (figures) data provided by the Bureau of Fisheries and Aquatic Resources (BFAR) ranking second only after Indonesia (Seafood TIP, 2020). This might suggest that the availability of disposed milkfish bones has also increased. Bones are a type of inedible food waste, as such, studies have been conducted with the aim to utilize them. There are studies that use milkfish bones as a calcium source in the form of flour to add calcium to the diet. In an Indonesian study by Darmawangsyah et al (2016), the results show that the calcium concentration of the pastry fortified with milkfish bone meal increases as more bone meal is added.

Milkfish bone waste can be used as an alternative source of calcium and phosphorus to provide a nutrient-rich food source. This is because the nutritional content of milkfish is not only in meat but also in the bones. Milkfish bones contain 4% calcium, 3% phosphorus and 32% protein (Sudoyo, 2009). This allows the potential for milkfish as an alternative foodstuff that can be processed into milkfish bone flour and can be added to food products such as krupuk. The use of fish flour as a substitute for wheat flour in puffed shrimp chips or krupuk is a promising alternative, especially in terms of the quality of nutrients produced. Utilization of fish bones can be a natural source of calcium for calcium as a food material and supplement. This will be a strategy to maximize the utilization of fish resources as well as to effectively reduce waste from the fishing industry.

Puffed shrimp chips, commonly known as krupuk, are a common snack in most Asian countries, including Philippines, Taiwan, and Thailand. Raw shrimp chips are traditionally manufactured by mixing shrimp, starch, and water to form a paste, which is steamed, cooled down, cut into chips, and sundried. Raw shrimp chips are quite hard and are smaller and darker than puffed shrimp chips. Raw shrimp chips have been traditionally puffed by deep-frying; however, microwavepuffed products are currently available in the market for health-conscious consumers (Wen-Chieh Sung,2015).

Moreover, the study will provided structure in using different types of methods in drying for making kroepoek out of milkfish bone as flour. The research aims to contribute to the development and usage due to the neglectance of potential using of milkfish bones as a resource especially in making flour. And one of the center of producing a numerous number of milkfish in Capiz is Pontevedra. In market demand also deboning of milkfish and throwing away the bones which is already waste to the community. The aim of this study is to carry out the usage of waste and to create a product in alternative way and to provide also a nutrient-rich food source that contains a high minerals.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The project aims to develop a product out of waste using bangus bone as replacement to the usual flour we used to put in cookies, crackers, and chips. This will also determine the acceptability of milkfish bone has potential to be used for flour substitution in making a kroepoek. There are studies conducted that utilize milkfish bones as a calcium source. Milkfish bone in the form of flour has been used as a fortification to add calcium and protein to food such as rice crackers (Eris et al, 2020; Jannah et al, 2020). The addition of milkfish bone into the products increased the calcium content. In a study by Darmawangsyah et al, 2016), the higher concentration of milkfish bone meal added in pastries increases the ash and calcium levels. This project is performed to establish the drying characteristics under different methods of locally manufactured crackers or kroepoek for future investigation of its final product quality in terms of proximate composition and different test. Physico-chemical teast, Physical characteristics,Sensory evaluation and products cost analysis will be assessed.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

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Objectives of the Study

This study will be conducted in order to determine the acceptability of Bangus bone in making Kroepoek.

Specifically, it aims to:

1. Determine the acceptability of Bangus bone on the sensory characteristics of Kroepoek in terms of:
 - 1.1 Taste
 - 1.2 Color
 - 1.3 Aroma
 - 1.4 Texture
 - 1.5 General Acceptability
2. Determine the physico-chemical characteristics of Bangus bone as Kroepoek in terms of:
 - 2.1 Expansion
 - 2.2 Mass
 - 2.3 PH level
 - 2.4 Moisture content
 - 2.5 Water activity
3. Evaluate if there is a significant difference on the acceptability of Bangus bone on the sensory characteristics of Kroepoek in terms of:
 - 3.1 Taste
 - 3.2 Color
 - 3.3 Aroma
 - 3.4 Texture
 - 3.5 General Acceptability
4. Evaluate if there is a significant difference on the physico-chemical characteristics of Bangus bone as Kroepoek in terms of:
 - 4.1 Expansion
 - 4.2 Mass
 - 4.3 PH level
 - 4.4 Moisture content
 - 4.5 Water activity
5. Conduct cost analysis of the Bangus Bone as Kroepoek

(11) Review of Literature/Prior Art Search (Related researches/activities that have been conducted. State-of –the-art of current technology/information from which the project proposal will take-off).

Closest Prior Art

CN103431450A PRODUCTION METHOD OF BAKED PUFFING SHRIMP CHIPS

Milkfish

Milkfish (*Chanos chanos* Forsskal, 1775) also locally known as “Bangus” is the prime fish commodity in the Philippines. It is the top fish commodity in terms of production and consumption. In 2020, milkfish production shared 416,315 metric tons (MT) or 17.9% of the total fisheries production that contributed about Php 43.5 billion to the Gross Domestic Product (GDP) of the country or approximately 2 to 3% of the country’s Gross Value Added (GVA) for Agriculture, Fishery and Forestry (AFF). Volume of milkfish harvested has increased from 225, 337 MT in 2001 to 416,315 MT in 2020 recording about 84.5% production growth for the past 20 years. On the average, its production performance marked a 3.37% average growth rate increment per year (PSA, 2020). About 10% of the 36.8 kg of annual fish consumed by an average Filipino household is milkfish (BFAR Philippine Fisheries Profile, 2018)

Fishbones in general have a high calcium and phosphorus content. In particular, they are rich in calcium carbonate making them a good alternative source of low-cost calcium carbonate, especially as they are generated

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in large amounts as solid waste from fish production activities (Corrêa&Holanda, 2019). The backbone of milkfish makes up 10-15% of the fish's whole-body weight (Wulandari&Kusumasari, 2019). Because of this, a practice called deboning in local fish markets. It is a tedious process wherein 170 intermuscular bones are removed. As a consequence, deboned milkfish or “boneless bangus” is a popular product of milkfish. This also leads to other processes that utilize the excess from trimming such as fishballs, milkfish lumpia, quekiam, embutido, and chicharon made from the skins (Yap et al, 2007).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories, and variables that indicate relevant relationships to be tested.)

An exploratory approach of using different methods was conducted to achieve the purpose of the study. To substitute the milkfish bone flour to the usual flour we use to attain the usage of waste product and obtain the right methods in creating a flour out of bones and to incorporate it as kroepoek to use the flavor also fo the milkfish bones. The different methods use in making flour out of milkfish bone is commonly used as the independent variable while parameters that are usually included to measure the acceptability and development of product are physical characteristics such as expansion and mass. In physicochemical charateristics are PH moisture and water activity. Finally, sensory evaluation, descriptive sesory profiling, cost analysis of the products are also being done to evaluate the product with milkfish bone as additional ingredients in making kroepoek.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

a. The protocol of the study is shown in the figure below:

Preparation of Ingredients and Materials

The ingredients will be get in the local market and/ or from suppliers within the vicinity. The following supplies and tools will be utilized in the study to prepare various treatments: bangus bone, cassava flour, and water. A working table, measuring cup, kitchen knife, chopping board, stove, mixing bowls, spoon, weighing scale, strainer, baking pan, plate, steamer, and frying pan will be utilized in the process.

Producing of Milkfish bone flour

Making of Kroepoek begins with the process of making bone flour. Method is adopted to the study of (Kusdarwati, 2021). First, milkfish bones were washed with water, then boiled for 30 minutes at a temperature of 100°C. Then, separate the excess meat from the bones. After that, the milkfish bones were heated in an autoclave using pressure cooker at a temperature of 121°C for 2 hours. Then dried using the three different methods (Sun-drying, Air-drying, and Microwave). Pulverized the milkfish bone and lastly Seiving to remove the bigger particles.

Making Kroepoek

In the preparation of the bangus bone as kroeppeck, all ingredients were combined in a mixing bowl. Base ingredients, bangus bone flour and water was added last. The mixture was placed on grease pan and form into logs. The logs were steamed for 30 minutes or until it becomes transluscent. After steaming, it was carefully removed, sliced and placed in a drying tray to cool down and air dried for about an hour. After an hour, the logs were sliced thinly, with about 1.5 to 2.0 mm thickness. It was again air dried for another 15 minutes before frying.

Experimental Design: One – factor Experiment in Completely Randomized Design (CRD)

Treatments:

A – Control (standard formulation)

- 225 base ingredients
- 15 ml water

B – Sun-drying method

- 225 grams base ingredients
- 100 grams bangus bone flour
- 15 ml water

C – Air-drying method

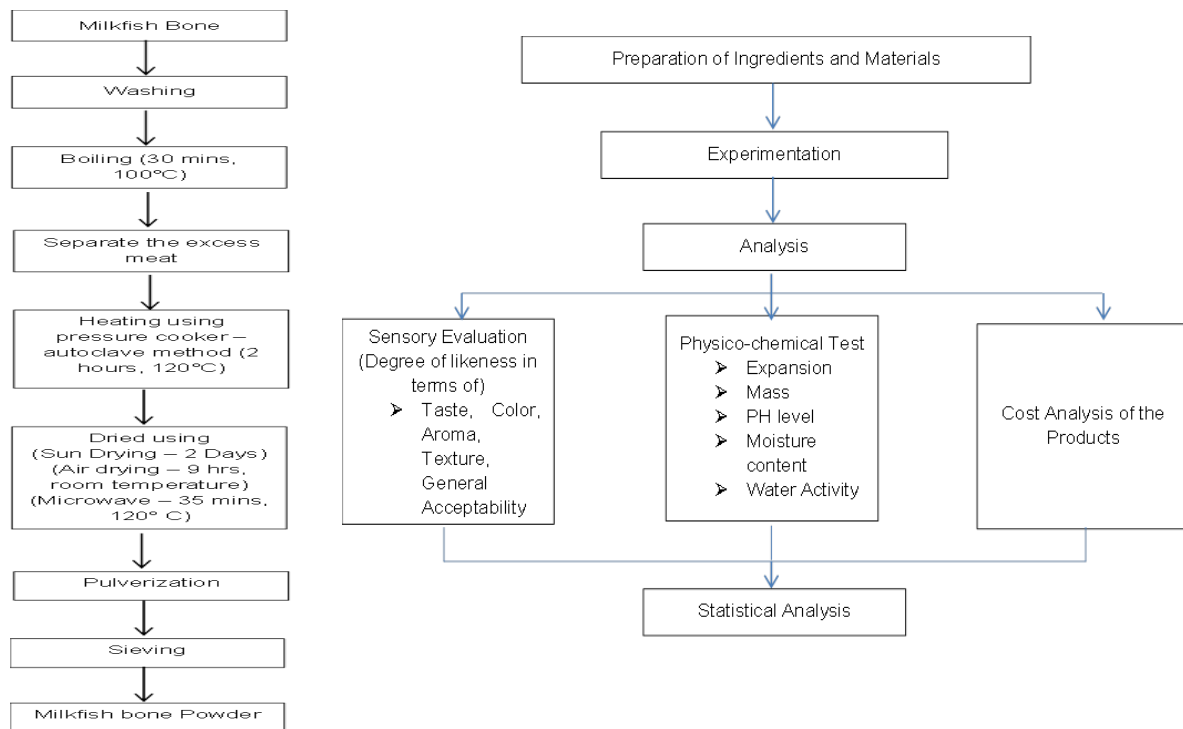
- 225 grams base ingredients
- 100 grams bangus bone flour
- 15 ml water

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D – Microwave method

- 225 grams base ingredients
- 100 grams bangus bone flour
- 15 ml water

Shelf life (in days) – 0, 3, 5 and 7 days old bangus bone flour



b. Experimental Lay-out

Table 1. The proposed experimental lay-out.

TREATMENT	SHELFLIFE (in days)			
	0	3	5	7
A – Control (std. formulation)				
B – Sun-drying method				
C – Air-drying method				
D – Microwave method				

Note: Experiment will be done in triplicate

c. Methods of Analysis

Physico-chemical analysis

The methods to be employed in the analysis will be based on the Association of Official Analytical Chemists (AOAC) Methods. The tests to be conducted will be as follows:

- 1.1 **Expansion.** A rape seed displacement method (Liu et al., 2010) was used to measure the volume of the kroepoek, and their expansion ratio was calculated using Eq. $\phi_1 \frac{V_2}{V_1}$ where . ϕ_1 is expansion ratio, V1 and V2 are volume of kroepoek before puffing and after puffing.
- 1.2 **Mass.** Measurement can be done by weighing the product developed per treatment.
- 1.3 **PH level.** For the pH determination, 1 gram of uncooked and 1 gram cooked of kropek samples

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dissolved in 10 ml water separately and analyzed using a pH pen (Apera Instruments).

- 1.4 **Moisture content.** Using oven (Mettler) set at $100 \pm 5^{\circ}\text{C}$, tared crucibles with an initial weight of approximately two (2) grams of uncooked kropek samples is dried until constant weights are achieved. Equation below is use for the computation of percent (%) moisture content (MC).

$$\% \text{ MC} = \frac{\text{Initial weight} - \text{Final weight}}{\text{Initial weight}} \times 100$$

- 1.5 **Water activity.** For the water activity (a_w) determination, crushed uncooked kropek was analyzed using a water activity meter (Novasina) until constant a_w values was obtained.

Selection of Panelist

This finished product will be evaluated as to color, taste, aroma, and texture and overall acceptability by the evaluators, composed of 5 faculty members and 5 students from Capiz State University- Pontevedra Campus.

Evaluation of the product

The acceptability of Bangus bone as Kroepoek will be evaluated by a panel of 10 members per replication. The treatment will be presented at random to the panel members to avoid positional bias. The tasters will be composed of selected students, employees, and faculty of Capiz State University, Pontevedra Campus. The qualities of the product will be evaluated by the panelists with the use of scorecards for sensor evaluation. Testing of the product in each replication will be done. The evaluators will be oriented as to how to evaluate the product. They will be guided in determining the acceptability of the different product using score cards containing a rating scale in the different sensor qualities such as color, taste, aroma, texture, and general acceptability.

Data Analysis

The scores given by the evaluators will be analyzed using the arithmetic mean and standard deviation. Appropriate test of difference will be employed to the data set; analysis of variance (ANOVA) will be used if the data is normally distributed while Kruskal-Wallis H Test will be used if the data does not satisfy the assumption of normality.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Activity	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Refinement of proposal						
Purchase and preparation of materials						
Experimentation						
Sensory Evaluation						
Data gathering						
Statistical analysis and interpretation						
Writing of report						

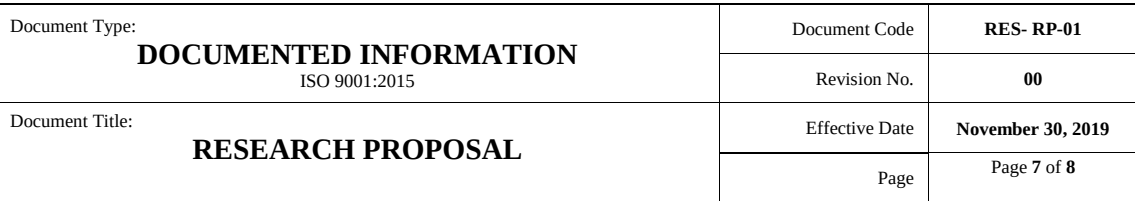
(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible.)

Publication

- at least 1 original article on the product development and testing
- at least 1 technical manual/guide
- at least 1 product specification

Patent/Intellectual Property

- at least 1 IP (Patent/UM) applications – method of preparation



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Sub-total for PS	11,068.00	11,068.00	11,068.00	11,068.00	11,068.00	11,068.00	66,408.00
II. MAINTENANCE AND OTHER OPERATING EXPENSES							
Travelling Expenses	500.00	500.00	500.00	500.00	500.00	500.00	3,000.00
Office supplies	5,000.00						5,000.00
Ingredients		10,000.00					10,000.00
Other Professional Expenses (Lab Analysis)				25,000.00			25,000.00
Other Professional Expenses (IP Protection)					15,000		15,000.00
Sub-total for MOOE	5,500.00	10,500	500.00	25,500.00	15,500.00	500.00	58,000.00
III. EQUIPMENT OUTLAY							
Microwave		5,000					5,000.00
Sub-total for EO		5,000.00					5,000.00
Grand Total	16,568.00	26,568.00	11,568.00	36,568.00	26,568.00	11,568.00	129,408.00

(19) Literature Cited (Include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Anggraeni, D., Am K., Eris, F.R., Munandar, A., (2020). Utilization of Milkfish (Chanos chanos) Bone Powder in Making of Rengginang, Local Food of Baduy Tribe. Department of Food Technology, Faculty of Agriculture, Universitas Sultan Ageng Tirtayasa Jalan Raya Jakarta Km 4 Pakupatan, Serang-Banten

Chen, T.Y., Hsu, P.H., Luo, H.M, Sung, W.C., (2015). Effects of calcium supplements on the quality and acrylamide content of puffed shrimp chips. Department of Food Science, National Taiwan Ocean University, Keelung, Taiwan

Dhahiyat, Y., Nuraeni, A., Pratama, R.I., Rostini, I., (2020). Milkfish bone flour fortification as a source of calcium on donut preference level. Global Scientific Journal, Vol. 8. , Issue 4.

Esma, G.T., Milloren, S.K., Muti, M.B, Nemeno, N.B., Tibe, A.O., (2023). Efficacy of Milkfish (Chanos Chanos) Bones as a Liquid Fertilizer on Plant Growth of Lettuce (Lactuca Sativa). International Journal of Research Publication and Reviews, Vol 4, no 3, pp 334-344

Kusdarwati, M.H.S. (2021). The Substitution Effect of Bone Fish Flour Milkfish on Physical and Chemical Characteristics of Cookies. Department of Fish Health Management and Aquaculture, Faculty of Fisheries and Marine, Universitas Airlangga, Kampus C Jalan Mulyorejo, Surabaya 60115 East Java, Indonesia

Kusumasari, S., Wulandari, P., (2019). Effect of extraction methods on the nutritional characteristics of milkfish (Chanos chanos Forsskal) bone powder. Department of Food Science, Faculty of Agriculture, University of Sultan Ageng Tirtayasa Jl. Raya Jakarta km. 04 Pakupatan, Serang City, Banten Province, Indonesia

(20) Capsule Curriculum Vitae (One-page brief C.V. only)